Confidential and verifiable upstream inference over standard TLS 1.3

A no-trusted-hardware protocol for LLM routing nodes

routstr-confidential project

2026-10-10

Abstract

LLM routing nodes forward user requests to upstream providers with the node's own API key, so users must trust the node with their prompts, and the node must keep its key from its users. Enclave-based designs ask every user to verify an attestation of the node's software after every update. Our protocol rests instead on the provider's TLS certificate and on cryptography. The client is the TLS 1.3 client of the provider and the node relays the records. Client and node run the TLS key schedule as a two-party computation (share conversion over P-256, then a dual-execution garbled circuit): the node alone gets the key of the first application epoch, in which it writes its `Authorization` record and a KeyUpdate, and the client alone gets the keys for the body and the streamed response. Nobody holds the ECDHE or master secret, so the client cannot recover the API key even if it records the node's traffic. Zero-knowledge proofs anchored on hashes the computation outputs show the client that the node's record matches a published template, and show the node that the request ends in a public suffix pinning the model and token cap and that the billed usage is the provider's own report. Against a live provider on one host, a cold request reaches the provider after 0.89 s and a pre-warmed one 5 ms after it is claimed, followed by a 0.10 s suffix proof. The price is about 21 MB of garbled tables in each direction per request, which pre-warming moves off the request path. We also describe a single-party variant that needs a tenth of the bandwidth but protects the key only while no observer of the node's egress colludes with the client.

Contents

1. Introduction **2**
1.1 Contributions . . . 3

2. Model **3**

3. Preliminaries and the provider profile **4**
3.1 Provider profile . . . 4
3.2 TLS 1.3 key schedule . . . 4
3.3 Cryptographic building blocks . . . 5

4. Protocol **5**
4.1 Who holds which secret . . . 5
4.2 Epoch structure . . . 5
4.3 Setup, preprocessing and the joint key share . . . 6
4.4 The two-party key schedule . . . 6
4.5 Credential record (epoch 0) and π_{NK} . . . 7
4.6 Body (epoch 1) and π_{C2S} . . . 8
4.7 Response and usage disclosure (π_{C3S}) . . . 8
4.8 Settlement . . . 9
4.9 Aborts . . . 9

5. Security analysis **10**

6. Implementation and evaluation **11**
6.1 Prototype . . . 11
6.2 Component costs . . . 11
6.3 End to end . . . 12
6.4 Over a network and bandwidth . . . 13
6.5 Live verification . . . 13

7. v1: the single-party variant **14**

8. Deployment pattern **15**

9. Related work and alternatives **15**

10. Conclusion **16**

Appendix A. Case study: the provider profile in practice **17**

References **17**

1. Introduction

Routing nodes and gateway services sit between users and upstream LLM APIs. A node holds an upstream account and forwards requests with its own API key; users pay the node (typically prepaid with e-cash tokens such as Cashu), and the node pays the upstream. Two trust problems follow.

The user trusts the node with everything. The node sees the prompt and the response; it can log, alter or replay them. This is a privacy problem (prompts contain source code, contracts, medical text) and a correctness problem (the node can substitute models or parameters).

The node must keep its credential from the user, and both must agree on the bill. The node's upstream API key must never leak to users. Charging must be defensible: the user should be able to check that the tokens billed are the tokens the upstream actually produced and reported.

The usual answer to the first problem is confidential computing: run the node, or the model, inside a trusted execution environment (TEE) and let users verify a remote attestation. That moves trust from the operator to the hardware vendor and to the measured software. For the guarantee to mean anything, a user has to know which software measurement is acceptable, which requires a reproducible build and a review of the whole stack that touches the plaintext, and has to repeat that after every update. In practice few users do this, and the attestation becomes a badge rather than a check.

This paper takes a narrower route. A user's request to a provider is already protected by TLS, and the provider is already authenticated by a WebPKI certificate. We **hand the TLS client role to the user and let the node relay ciphertext**. The node contributes its credential to the request without the user learning it, and cryptographic proofs pin the few bytes each side may write and bind billing to the provider's own numbers. There is no attested software to track: the user checks a certificate and a handful of proofs whose statements are fixed by the protocol.

The difficulty is the credential. If the user's client runs the TLS handshake alone, it knows every traffic secret, including the key of the record that carries the node's API key; the key is then hidden only as long as the client never sees that record's ciphertext. We remove that condition by computing the handshake key schedule jointly: the node obtains the key for its own record, the client obtains the keys for the body and the response, and neither can derive the other's. The two-party computation (2PC) covers only the key schedule, a fixed workload of 31 SHA-256 compressions, never the payload.

1.1 Contributions

1. A TLS 1.3 protocol (section 4) in which the first application epoch belongs to the relay node, which writes its credential record and a KeyUpdate under a key only it holds, while the body and the response run under keys only the client holds. The handshake key schedule runs as a two-party computation with security against a malicious peer and at most one bit of leakage per session (section 4.4).
2. A request-integrity construction: the client proves that its request body ends in an agreed public JSON suffix that pins every cost-relevant parameter, and the provider sees the complete request only after the proof verifies (section 4.6). The node proves that its hidden record matches a published template (section 4.5).
3. A verifiable-accounting construction: the client proves that the upstream's terminal usage event is authentic decrypted output of the session, anchored at the end of the relayed transcript, and the node charges exactly that usage at the rates of the offer the client verified, through its ordinary billing (sections 4.7, 4.8). Proofs that decrypt reveal one bit (equal or not), never decrypted bytes.
4. A **provider profile** (section 3.1) that separates what any upstream must satisfy from provider-specific details.
5. A working implementation verified end to end against a live provider, with measurements of every component and of the whole path, cold and pre-warmed (section 6), and a comparison with a simpler single-party variant that trades the credential guarantee for bandwidth (section 7).

2. Model

Parties:

- **C, client** (a user's daemon or app). TLS 1.3 client of the upstream session; the only holder of the body and response traffic secrets; payer of the node.
- **N, node.** Holds the upstream API key; owns the TCP connection to the upstream and relays it; the only holder of the first application epoch's key; bills C through its ordinary accounts.
- **U, upstream** (any provider whose profile, section 3.1, is satisfied). Unmodified.

Channels: C to N is a control channel, a byte relay and one channel per proof or 2PC (in our deployment, WebSockets); N to U is ordinary TCP:443.

Goals:

ID	Property
P1	Credential confidentiality: only N knows the upstream API key. C learns nothing about it even if it sees every byte N sends to U.
P2	Content confidentiality: only C (and U) can read the prompt and the response. N sees ciphertext plus the public billing data: the usage event, or on an upstream error the response head.
P3	Integrity: C can verify that the request U received is C's own content behind the agreed public template content (not its exact wire bytes, residual risk 1), and that the response is exactly what U sent (tool calls included).
P4	Verifiable accounting: the charge is computed from the token usage the upstream itself reported in the session, verified by N (who settles) and by C (who checks the signed receipt against its own plaintext).
P5	Upstream authenticity: both parties verify that the session terminates at a server holding a valid certificate for the upstream hostname.
P6	Freshness: every session is live and bound to one payment; nothing can be reused across sessions.

Non-goals: hiding the prompt from U (U processes it by definition); proving which model weights ran; hiding traffic volume and timing from N (section 5); and third-party verifiability. The proofs convince the two parties of the session, which is what settlement needs. They are interactive and designated-verifier, and C, holding the server's application key, could forge server records for anyone but N. A receipt is N's signed statement, not a proof to outsiders.

Adversaries: either C or N may deviate arbitrarily from the protocol (they are malicious towards each other); U follows TLS and its own API, and is trusted with the prompt and the API key as it is today. Assumptions: the standard hardness

of ECDH over P-256, SHA-256, HMAC and AES-GCM; security of the garbling scheme, of oblivious transfer and of VOLE-based zero knowledge; the WebPKI. No trusted hardware.

3. Preliminaries and the provider profile

3.1 Provider profile

The protocol is defined for any upstream satisfying:

1. **TLS 1.3** with an AES-GCM suite in the SHA-256 family (this work: `TLS_AES_128_GCM_SHA256`) and ECDHE over **secp256r1**. No compression, no 0-RTT, no HelloRetryRequest for a P-256 share.
2. **HTTP/1.1** with explicit `Connection: close` and `Accept-Encoding: identity` (record-level epoch interleaving requires HTTP/1.1 framing; HTTP/2 would need a different record mapping).
3. **KeyUpdate accepted mid-request** (client-initiated, `update_not_requested`).
4. **No processing before the complete body.** The provider starts work only once the whole `Content-Length` body has arrived. N relies on this to stop a request whose suffix does not verify (section 4.6).
5. **Parameter pinning semantics**: a JSON request whose *effective* parameters are fully determined by a public suffix that the client can append. The provider must:
   - keep the last value of duplicate top-level keys;
   - let a later object value replace an earlier one rather than merge with it;
   - reject malformed JSON and unknown fields, so the private prefix cannot leave an object or a string open for the suffix to land in.
6. **Cost-relevant parameters overridable by the suffix**: model, token caps (which must cap reasoning tokens too), sampling count, and provider feature flags that change price, including any injected system prompt. The provider must reject any other price-changing field. The size of the prompt needs no bound, because the node reserves for a full context window (section 4.6).
7. **A streaming (or terminal) usage event** carrying input/output token counts and, ideally, a provider-computed cost figure, emitted before the response ends.
8. **No reflection of the request head into the response.** C reads the response, so a provider that echoed the `Authorization` header (for example in an error body) would hand C the key.

Items (5) and (6) quantify over every field and input, which no finite test can establish. For the case-study provider (Appendix A) a live regression gate tests items 1 to 7 with sampled cases; the requirements hold as far as those samples show, and the gate is re-run after provider changes. Item 8 is a property of the provider's error handling that the gate does not test. Everything except the specifics of items 4 to 7 is provider-generic: the epoch structure, the 2PC, the proofs and the record assembly do not change per provider; only the offer (template, pinned keys, price list, usage parser) does.

3.2 TLS 1.3 key schedule

The TLS 1.3 key schedule (RFC 8446 section 7.1, SHA-256) derives, from the ECDHE shared secret Z and the transcript hashes th_SH (through ServerHello) and th_SF (through server Finished):

$$HS = \text{HKDF-Extract}(\text{Derive-Secret}(ES, \text{"derived"}), Z)$$

$$c_{hs}, s_{hs} = \text{HKDF-Expand-Label}(HS, \text{"c hs traffic"} \,/\, \text{"s hs traffic"}, th_{SH})$$

$$MS = \text{HKDF-Extract}(\text{Derive-Secret}(HS, \text{"derived"}), 0^{256})$$

$$c_{ap,0}, s_{ap,0} = \text{HKDF-Expand-Label}(MS, \text{"c ap traffic"} \,/\, \text{"s ap traffic"}, th_{SF})$$

$$c_{ap,i+1} = \text{HKDF-Expand-Label}(c_{ap,i}, \text{"traffic upd"})$$

where ES is a public constant without a PSK. Each traffic secret yields an AES key K and IV through the "key" and "iv" labels. A record with sequence number n uses nonce $IV \oplus n$; its ciphertext is the plaintext XOR the AES-CTR keystream from counter 2, and its tag is $\text{GHASH}_H(AAD, C) \oplus \text{AES}_K(J_0)$ with $H = \text{AES}_K(0^{128})$. Every step is a

one-way HMAC-SHA-256, so any set of secrets on the right of a derivation reveals nothing about the secrets that are not derived from it.

KeyUpdate: a sender can rotate its own sending keys at any record boundary ($c_{ap,i} \to c_{ap,i+1}$); the receiver updates likewise. Sequence numbers restart in each epoch.

3.3 Cryptographic building blocks

VOLE-based zero knowledge. We use the interactive QuickSilver-style ZK of TLSNotary's `mpz2`, where SHA-256 (as a compression-function circuit) and AES-128 are available as gadgets. Both parties are online during the session, so interactivity costs only round trips. Every plaintext comparison in our statements is done inside the circuit and only a single equality bit is decoded.

Garbled circuits and dual execution. A garbler encodes a Boolean circuit so that an evaluator who holds one label per input wire learns the output and nothing else. Half-gates garbling costs 32 bytes per AND gate (XOR is free). Plain garbling is secure only against a semi-honest garbler. *Dual execution* (Mohassel and Franklin 2006; Huang, Katz and Evans 2012) runs the circuit twice with the roles swapped and checks that both runs produced the same output, which gives security against a malicious party at the cost of at most one bit of leakage: the result of the check.

Share conversion (ECtF). Two parties holding ECDH contributions can convert the two halves of a point sum into additive shares of its x-coordinate with a few multiplicative-to-additive (M2A) and additive-to-multiplicative (A2M) conversions over the base field, built on oblivious linear evaluation. TLSNotary uses the same construction for its key exchange.

4. Protocol

4.1 Who holds which secret

The protocol is built around a split of the TLS secrets. C and N jointly emulate one TLS client; the 2PC releases to each party only what it needs:

Value	C	N	Purpose
ECDHE scalar, Z	additive share	additive share	key exchange
HS, MS, $c_{ap,0}$	never	never	inside the 2PC only
c_{hs}, s_{hs}	yes	yes	both verify the handshake (P5)
K_0, IV_0 (from $c_{ap,0}$)	**never**	yes	N's credential record (P1)
$c_{ap,1}$	yes	**never**	C's request body (P2)
$s_{ap,0}$	yes	**never**	the response (P2)
$h_1 = \text{SHA256}(c_{ap,1})$, $h_2 = \text{SHA256}(s_{ap,0})$, $h_k = \text{SHA256}(K_0 \| IV_0)$	yes	yes	public anchors for the proofs

$c_{ap,1}$ is derived from $c_{ap,0}$, but not the other way round, so C's body key does not reveal N's record key. Neither K_0 nor IV_0 yields $c_{ap,0}$. U sees an ordinary TLS client.

4.2 Epoch structure

Epoch	Key	Writer	Content
Handshake	c_{hs} / s_{hs}	C (U)	ClientHello with the joint key share; client Finished computed by C from c_{hs}

Epoch	Key	Writer	Content
0	$c_{ap,0}$ (K_0, IV_0)	**N**	sequence 0: the credential record (request head with `Authorization`); sequence 1: `KeyUpdate(update_not_requested)`
1	$c_{ap,1}$	C	the JSON request body; its final record is the public suffix, released to U by N only after π_{C2S}
Response	$s_{ap,0}$	U	streamed response; N relays ciphertext

C writes nothing in epoch 0 and starts epoch 1 at sequence 0. N writes exactly two records, atomically. U processes epoch 0 as an ordinary client's first records followed by a KeyUpdate, and decrypts everything after it under $c_{ap,1}$.

4.3 Setup, preprocessing and the joint key share

Setup. C opens a control channel to N and sends its ordinary node credential, the signature of the offer it verified, the model, the token cap and the body length. N validates the credential, freezes the offer's rates, reserves funds for the request (section 4.6) and only then dials its relay process. N opens no connection to U before a valid setup.

Preprocessing. Before any TLS byte is sent, both parties garble both phases of the key-schedule circuit (section 4.4) with their own global offsets and send each other the tables, and run the oblivious-transfer setup for every evaluator input bit and the random OLEs for the share conversion. This step is independent of all inputs. It moves about 21 MB in each direction, because the garbled tables cannot be reused: a garbled circuit is single-use, and each session needs one handshake and one credential record. Section 6 discusses how pre-warming moves it off the request path.

Joint key share. C draws a scalar a_C, sends a commitment SHA256$(P_C \| \rho)$ to $P_C = a_C G$, receives $P_N = a_N G$ from N and opens. Both compute $P = P_C + P_N$. Committing first prevents C from choosing $P_C = T - P_N$ for a point T whose discrete logarithm it knows. Scalars are fresh in every session. The key exchange uses P-256 rather than X25519 because the joint share needs full point addition, which X25519's x-only Montgomery form does not offer.

ClientHello enforcement. C builds a ClientHello with exactly one key share, the joint point P, on secp256r1; `supported_groups` lists only secp256r1; one cipher suite; no PSK and no early data. **N parses the ClientHello and relays it only if all of this holds.** Without this check C could add a share of its own, finish the handshake alone and learn every secret. N rejects a HelloRetryRequest and checks that the ServerHello selects secp256r1 and the offered suite, then extracts U's share S.

4.4 The two-party key schedule

ECtF. C computes $a_C S$ and N computes $a_N S$. A point-addition share conversion (two A2M and one M2A over the P-256 base field) turns these into additive shares $x_C + x_N = x(a_C S + a_N S) \bmod p$ of the ECDHE secret Z, without either party learning Z.

Phase A (after the ServerHello). A garbled circuit takes x_C, x_N, th_SH and both parties' random masks. It adds the shares modulo p (about 770 AND gates), computes HS, its keyed HMAC states, c_{hs}, s_{hs}, the "derived" secret and MS. It outputs c_{hs} and s_{hs} in the clear and MS only as $MS \oplus r_C \oplus r_N$, so that afterwards C's share of MS is r_C and N's is $MS \oplus r_C$. Neither share says anything about MS. Phase A must finish before U gives up waiting for the client Finished; locally it takes 23 ms.

Handshake verification. With s_{hs}, C decrypts U's encrypted handshake messages in its own TLS stack and verifies the certificate chain to the WebPKI roots, the hostname, CertificateVerify and the server Finished, then computes and sends its Finished from c_{hs}. N independently decrypts the same messages, reconstructs the transcript and verifies

the chain, hostname, CertificateVerify and both Finished messages, fail-closed (P5). A wrong Z, for example after a deviation in ECtF, fails the server Finished here.

Phase B (after the server Finished). A second circuit takes the MS shares, th_SF and fresh masks, and computes $c_{ap,0}$, $s_{ap,0}$, $c_{ap,1}$, the record key and IV of $c_{ap,0}$, and the three anchors. It outputs $(s_{ap,0} \| c_{ap,1}) \oplus m_C$, which only C can unmask, $(K_0 \| IV_0) \oplus m_N$, which only N can unmask, and h_1, h_2, h_k in the clear.

Every HKDF-Expand-Label in the schedule fits one inner block, and keyed HMAC states are computed once per secret and reused, so phase A costs 14 SHA-256 compressions (298,088 AND gates) and phase B 17 (358,091 AND gates).

Dual execution. Each phase is garbled twice: in run 1 N garbles and C evaluates, in run 2 C garbles and N evaluates. Both runs execute at once. Every output wire is decoded by both evaluators, which is why private outputs are masked inside the circuit with a mask only their owner inputs: the other party decodes a one-time-padded value. Each party also inputs its own view of the transcript hash, so a disagreement about the transcript changes the output and fails the check below.

Equality check. Before either party uses any output, they compare the two runs through labels. C hashes, for every output bit i, the label it obtained in N's circuit for the value v_i it decoded, together with its own run-2 label for that same value; N does the same with the value it decoded in run 2. The hashes are exchanged commit-then-open. A party can produce the other's hash only if both runs decoded the same vector: for a differing bit it would need an inactive label of the peer's circuit, which half-gates garbling hides. On a mismatch the session aborts. **No output is sent to U, used to encrypt, or revealed before the check passes.**

What a cheating party gains. A corrupt garbler (wrong circuit, wrong decoding bits, inputs that differ between the two runs) and selective failure on the oblivious transfer can only make the honest party's decoded output a wrong function of the honest inputs. The adversary learns whether the check passed, one predicate on those inputs, and the session aborts. The honest inputs are the ECDHE share, the masks and the MS share, all fresh in each session. The API key is never an input to the 2PC, so leakage cannot accumulate on it across sessions. A party that feeds different MS shares or masks to phase B than phase A gave it only obtains keys nobody knows, which U rejects. ECtF uses semi-honest conversions; a deviation there only shifts Z, which the server Finished check catches, or causes a selective failure that again surfaces as one success-or-abort bit.

Inputs are never opened. Every private input enters only as a garbled input or an OLE input, and nothing is revealed after the session. This distinguishes the construction from 2PC-TLS designs in which one party opens its shares afterwards (section 9), which N cannot do here: its shares would give C the master secret and therefore K_0.

What each party learns, in total:

Party	Learns	Does not learn
C	c_{hs}, s_{hs}, $s_{ap,0}$, $c_{ap,1}$, h_1, h_2, h_k; whether a check passed	Z, HS, MS, $c_{ap,0}$, K_0, IV_0, the API key
N	c_{hs}, s_{hs}, K_0, IV_0, h_1, h_2, h_k; whether a check passed; the suffix and the usage event (as one-bit-checked public values)	Z, HS, MS, $s_{ap,0}$, $c_{ap,1}$, prompt, response
U	an ordinary TLS 1.3 session	that two parties are involved

4.5 Credential record (epoch 0) and π_{NK}

After phase B and the handshake verification, N fills the published head template with its API key and C's declared body length, encrypts it under K_0 at sequence 0, and seals `KeyUpdate` at sequence 1. It sends C a commitment $c = \text{SHA256}(header \| C_1)$ to the ciphertext and proves its content.

π_{NK} (N to C).

- *private*: K_0, IV_0, the key bytes;
- *public*: h_k, the template with the key positions, the key alphabet, $|head|$, the record header, the commitment c;

- *statement*: SHA256($K_0 \| IV_0$) $= h_k$; the AES-CTR keystream under K_0 with nonce $IV_0 \oplus 0$; $C_1 = (head \| \texttt{0x17}) \oplus$ keystream; SHA256($header \| C_1$) $= c$; $head$ equals the template outside the key positions; every key byte is in the alphabet. Only the digests and the alphabet bit are decoded.

C rebuilds every public input from the signed offer, its own body length and the h_k that its own 2PC output, so N cannot prove a different template or a key other than the one the 2PC produced. The proof binds the head's content and length. It does not bind the bytes N actually writes: C never sees C_1, so a different head of the same length is not detectable by C (section 5, residual risk 1).

When π_{NK} has verified, C claims the session. N writes the credential record and the KeyUpdate in one write; C's epoch-1 records, held by N until then, follow. There is no client proof in this step: N never handed out anything that C must prove correct.

4.6 Body (epoch 1) and π_{C2S}

C writes body = body_private ‖ suffix under $c_{ap,1}$ from sequence 0, with $|body|$ fixed at setup. body_private opens the JSON object and may carry messages, tools and other schema-valid fields. The suffix is the public string that pins every cost-relevant parameter: model, token caps, `n`, streaming, usage reporting and provider feature flags. N's relay builds it from the model and token cap declared at setup and accepts no other. The provider resolves the effective values to the suffix (section 3.1, item 5).

C sends the suffix as its own final record. N forwards the earlier body records but **holds the record that completes the body** until π_{C2S} verifies. N enforces that the epoch-1 plaintext is exactly $|body|$ bytes, so that record is identifiable. Until it is released U has an incomplete request and does nothing (section 3.1, item 4). If the proof fails, N closes the connection: no inference ran and nothing is billed. A client therefore cannot run a different model or a larger cap than N reserved for.

π_{C2S} **(C to N, before U can start).**

- *private*: $c_{ap,1}$;
- *public*: h_1, the record's sequence number, a ciphertext window of the final record and its offsets, the suffix;
- *statement*: SHA256($c_{ap,1}$) $= h_1$; derive K_1, IV_1; CTR-decrypt the window; its end equals the suffix. The verifier decodes one bit.

Because the anchor is a hash of the secret that the 2PC itself output to both parties, the statement does not recompute the key schedule: it costs 7 SHA-256 compressions and 6 AES blocks, 0.10 s locally.

Parameter validation by C. The public parameters of π_{C2S} and π_{C3S} come from N. C checks each against its own view before proving: π_{C2S} must be a window ending at the end of C's own final body record (sequence number, offset and ciphertext bytes as C sent them) with C's suffix as the expected value; π_{C3S} (section 4.7) must cover the final server records exactly as C received them, matched from the end of the transcript with at most one trailing close_notify record, with C's own disclosure as the expected value; the anchors must equal C's own 2PC outputs. N therefore cannot aim a proof, or its one-bit result, at other plaintext.

Reservation. The prompt is opaque to N, so the reservation never discounts it: it is max(m, $\lceil (L \cdot r_{in} + T \cdot r_{out})/1000 \rceil$) msat, where L is the model's prompt limit (context length minus its completion limit), T the pinned token cap, r_{in}, r_{out} the per-1k rates of the frozen offer and m the node's minimum charge. For the case study (`deepseek-v4-flash`, L = 104,858, T = 40) this reserves 2,282 msat; the session cost 63.

4.7 Response and usage disclosure (π_{C3S})

U's response streams under $s_{ap,0}$; C decrypts it and verifies every GCM tag natively (P3 for the response) and streams tokens to the application live. At close, N signs a **session attestation** binding the session id, the certificate chain hash, the handshake transcript hash, directional ciphertext hashes, record boundaries and time.

C then discloses R: the plaintext from the boundary of the last usage-bearing data event through the stream terminator, with its chunk framing, anchored at the end of the transcript N relayed.

π_{C3S} **(C to N).**

- *private*: $s_{ap,0}$;
- *public*: h_2, the final records' ciphertexts and sequence numbers, R, and a count `skip` of undisclosed bytes after R;
- *statement*: SHA256$(s_{ap,0}) = h_2$; derive the server key and IV; CTR-decrypt the covered records; R ends `skip` bytes before the end of the response; in-circuit equality with R, decoded as one bit. For a usage disclosure `skip` is 0.

The choice of R is unambiguous: JSON strings cannot contain raw CR/LF, so event boundaries and HTTP chunk framing cannot be forged from model output. We checked this by prompting the model to emit fake usage events; every imitation stayed escaped inside a string (Appendix A).

4.8 Settlement

Confidential sessions use the node's ordinary billing; nothing about money is specific to the protocol:

1. **Setup.** C authenticates with the credential it already uses for normal requests, either an API key or a one-off e-cash token that the node turns into one, and names the offer it verified by its signature. N keeps the offers it signed in the last ten minutes and freezes that offer's per-model rates for the session; an unknown or expired offer is refused (HTTP 409) and C refetches it. N reserves on the credential (section 4.6) before any connection to U is opened.
2. **Charge.** N's relay parses R (dechunk, take the usage event and the terminal marker) and requires the event's model to equal the setup model; N requires it again before charging. N charges exactly the usage at the frozen rates, $\lceil (r_{in}(p-c) + r_{cached}\min(c,p) + r_{out}\cdot o)/1000 \rceil$ msat for p prompt tokens of which c cached and o completion tokens, capped at the reservation, through the node's ordinary finalize path.
3. **Receipt.** N signs (BIP-340 Schnorr, over canonical JSON) a receipt with the session id, model, usage, cost, reservation, the resulting balance, the frozen rates, its key, the SHA-256 of the exact attestation bytes, and the time. C requires the signature, takes every field from the signed bytes, and checks the session id and attestation hash against its own session, the usage against the plaintext it already has, and the cost against that usage at the signed rates. Unused balance stays on the credential and is withdrawn through the node's normal refund endpoint.

If C does not disclose after U received the request, N charges the full reservation, which is why disclosure is always in C's interest.

Upstream errors. An HTTP-level error from U (for example 503, 429, or a 400 for a request U rejects) carries no usage event. C then discloses only the response *head*: the status line, the header lines and the blank line that ends them. π_{C3S} proves it with `skip` set to the length of the rest of the response, so the head is anchored at a known distance from the end of the transcript and the error body, which can quote the request, stays private. For statuses the provider answers before doing any work (4xx, 503), N releases the reservation; other 5xx are charged like a missing disclosure. The head is unambiguous for the same reason as the usage event: a strict status line followed by well-formed `Name: value` header lines cannot come from model output, nor from chunk framing.

4.9 Aborts

Situation	Outcome
Any 2PC check fails, the ClientHello is rejected, or the handshake does not verify	session discarded before anything is written under K_0; reservation released
Abort before N releases the final body record (including a failed π_{C2S})	U never received a complete request; reservation released
Usage disclosed	charged per section 4.8
U returned a 4xx or 503 (disclosed)	reservation released
Released, no usage disclosed (or another 5xx)	full reservation charged; C keeps its transcript and attestation

5. Security analysis

P1 (credential). The API key exists in the protocol only as plaintext of N's epoch-0 record, encrypted under K_0. C never obtains K_0, IV_0, or any secret from which they can be derived (Z, HS, MS, $c_{ap,0}$): the 2PC outputs these to nobody or to N alone, masked under N's mask, and nothing is opened afterwards. The secrets C does receive (c_{hs}, s_{hs}, $s_{ap,0}$, $c_{ap,1}$) are one-way HKDF outputs that are not ancestors of K_0, and h_k is a hash of a 28-byte uniformly random value. So a C that records every byte N sends to U, alone or with an observer on N's network path, holds AES-GCM ciphertext under a key it does not know. A cheating C can learn at most one bit about its session's fresh shares (section 4.4), and the API key is not a 2PC input at all. C cannot cut N out of the key exchange: N relays only a ClientHello whose single key share is the joint point. The commitment to C_1 that N sends is also safe: without K_0 it does not allow testing guesses of the key.

P1 still assumes that U does not send the key back to C (profile item 8), and that N's own process and U are not compromised.

P2 (content). The body runs under $c_{ap,1}$ and the response under $s_{ap,0}$, which the 2PC outputs only to C, masked with C's mask. N's outputs (c_{hs}, s_{hs}, K_0, IV_0, the anchors) are not ancestors of either. A cheating N that garbles a wrong circuit, for example one that outputs C's keys in a form N can read, is caught by the equality check before C uses any key, and learns at most one bit about C's fresh inputs. The proofs that decrypt under C's keys (π_{C2S}, π_{C3S}) output one equality bit, on parameters C has checked against its own records, so N cannot use them as a decryption oracle. What N learns in plaintext is public by design: the suffix, the usage event, or on an error the response head.

P3 (integrity). C's body records are authenticated under a key only C has. N holds K_0 and can write any epoch-0 records (below), but none of them carries C's data. The suffix is proven to be the body's tail, so the effective parameters are exactly the agreed ones; response tags are verified by C.

P4 (accounting). The billed usage is a substring of the authenticated response, proven under the session's server secret and anchored to the end of the transcript. The anchors h_1, h_2 are outputs of the same 2PC that produced the keys, agreed by both runs, so C cannot prove against a secret of its choosing without a SHA-256 preimage. N computes the suffix from the setup values, accepts only that suffix, and lets U see the complete request only after π_{C2S} has verified, so U runs exactly the reserved model and cap. The receipt binds the result to the session id and the reservation.

P5 (authenticity). Both parties verify the certificate chain, hostname, CertificateVerify and both Finished messages: C in its TLS stack, N from the handshake secrets that the 2PC gives it in the clear.

P6 (freshness). The session id is a hash of fresh nonces and the node's one-time reservation id; the TLS randoms, the ECDHE scalars, the masks and all garbled circuits are fresh per session.

Residual risks.

1. *N controls epoch 0.* N holds K_0, so it can write any number of records in epoch 0, not only the proven head, and π_{NK} binds the head's content, not the bytes N writes. N could, for example, use another account or endpoint, or add a header that enables provider-side retention. C cannot see N's wire bytes and cannot detect this. The effect is bounded: epoch 0 carries none of C's data, C's own records stay authenticated under $c_{ap,1}$, the response is readable only by C, and a broken request shows up as a failed response. A node could always choose to misuse its own account; what the protocol guarantees is that it cannot read or alter C's content.
2. *The 2PC is new code.* The ECtF, the dual-execution garbling and the equality check are our own composition of `mpz2` components and have not been independently reviewed. The single-party variant (section 7) avoids this code.
3. *Selective failure.* Each 2PC session can leak one bit about the honest party's fresh shares through the abort decision. The bits do not accumulate on any long-term secret. A party that aborts sessions on purpose is visible to the other side.
4. *P-256.* The key exchange uses P-256, as most TLS clients still offer; X25519 cannot be used with an additive joint share.
5. *Traffic analysis.* N sees the size of every record. The body length is visible, and so is the response's token-length sequence, because streaming providers emit one record per token or small token group. This side channel has been shown to reconstruct a large share of LLM responses (Weiss et al., USENIX Security 2024). Compared with a node that sees the plaintext it is strictly less exposure; compared with "ciphertext only", it is more than

volume and timing. The prompt side can be padded inside the JSON body; the response side needs provider-side padding or batching.

6. *DoS and limits.* The node and its relay validate the setup and reserve before opening any connection to U, bound concurrent sessions (32), the time to setup (15 s), the session lifetime (600 s), the 2PC (180 s) and the body (8 MiB). The 2PC's preprocessing makes a session expensive for N in bandwidth (about 21 MB of upload each), which a reservation per session limits but per-client rate limits must also bound.
7. *Error metadata.* On an upstream error N learns the status line, the response headers and the body length, but not the body.
8. *Endpoint hygiene.* Keys live in C's process; N holds the API key in memory. C's certificate validation is RFC 5280 path validation to the WebPKI roots without revocation checks.
9. *Interactivity.* There is no artifact a third party can check. C holds $s_{ap,0}$ and could forge server records, so a portable artifact would need a design in which no single party holds the server key.
10. *Provider drift.* A provider change that breaks parameter pinning or usage reporting turns a verified session into an unverifiable one, which clients treat as failure, not fallback. The profile gate must be re-run per provider.
11. *Balances after an abort.* If N cuts the stream before the usage event, it charges the reservation, and C cannot prove who aborted. This is the trust the existing Routstr design already places in the node; in the honest case the cost is computed from upstream-authenticated usage, unbilled errors and sessions that never reached U cost nothing, and the receipt binds all of it. The amount at N's discretion is the reservation for the requested model.

6. Implementation and evaluation

6.1 Prototype

The implementation consists of:

- `cu-crypto`: native TLS 1.3 key schedule and GCM, tested against RFC 8448 and NIST vectors.
- `twopc`: the two-party key schedule (ECtF over P-256, the phase A and B circuits, dual execution over `mpz2`'s half-gates garbler with KOS oblivious transfer, the label equality check), with a client-side binary.
- `zkbench`: the VOLE-ZK statements π_{NK}, π_{C2S}, π_{C3S} (and those of the single-party variant) with tests that the circuits agree with native code, mutation tests, and one-bit disclosure tests; the client's prover binary reads its job and witness on stdin, so no secret touches the disk.
- `cu-sidecar`: N's relay (Rust). It enforces the ClientHello and ServerHello rules, runs N's side of the 2PC, verifies the handshake fail-closed, assembles and writes the epoch-0 records, holds the final body record, and verifies C's proofs.
- A reference TypeScript client, and the production path: a transport in the Routstr client SDK used by the client daemon, and a WebSocket route in the node that relays to the sidecar and bills through the node's ordinary accounts.

C's TLS stack is a fork of an MIT-licensed TLS 1.3 client with two hooks: an external key share, and traffic secrets injected from the 2PC instead of computed locally. It accepts only the version, suite and group it offered, validates certificate paths per RFC 5280 and hostnames per RFC 6125, and uses the full 64-bit sequence number in the nonce.

6.2 Component costs

All component figures are from an Apple M2 Max, release builds, both parties in one process. "Local" has no delay; "50 ms RTT" adds 25 ms one way and limits each direction to the stated bandwidth.

Two-party key schedule.

Circuit	AND gates	XOR gates	SHA-256 compressions	Output bits
Phase A	298,088	1,421,372	14	768
Phase B	358,091	1,705,600	17	1,504

Each circuit is garbled twice, so a session garbles 1.31 M AND gates, half by each party. Only 8 of phase A's compressions are on the path to c_{hs} and s_{hs}.

Step (local)	Time	Data
Preprocess (garbled tables, OT setup, OLEs)	≈0.49 s in the live client (126 ms in-process)	≈21 MB each way
Joint key share	3 ms	≈115 B
ECtF	2 ms	≈0.3 kB
Phase A / its check	23 ms / 4 ms	≈21 kB each way
Phase B / its check	26 ms / 7 ms	≈24 kB each way

Online, the whole 2PC moves about 45 kB in each direction. Preprocessing is bandwidth-bound: at 50 ms RTT it takes 1.8 s at 100 Mbit/s and 8.6 s at 20 Mbit/s. Measured against a deployed node from a home connection, it took 24.6 s; the rest of that session (online 2PC about 0.5 s, proofs about 3 s) was small by comparison, so on consumer uplinks pre-warming is what makes v2 usable. The online steps take about 0.45 s at 50 ms RTT (3.5 round trips plus about 30 ms of computation after the ServerHello, 2 round trips after the server Finished).

Proofs. The anchors remove the key schedule from the client's proofs, and π_{NK} replaces the single-party variant's pair $\pi_{C1} + \pi_N$ (section 7):

Proof	Local	50 ms RTT	Data	SHA-256 compr.	AES blocks
π_{NK} (credential record)	0.17 s	1.17 s	1.45 MB	5	15
π_{C2S} (suffix)	0.10 s	0.79 s	0.96 MB	7	6
π_{C3S} (usage)	0.22 s	1.22 s	1.47 MB	7	38
v1 π_{C1} (record material)	0.46 s	1.51 s	1.66 MB	32	18
v1 π_N (head template)	0.08 s	0.79 s	0.95 MB	4	0
v1 π_{C2} (suffix)	0.45 s	1.51 s	1.67 MB	36	6
v1 π_{C3} (usage)	0.46 s	1.53 s	1.67 MB	28	38

The 50 ms RTT column uses 100 Mbit/s. At 50 ms RTT about 0.7 s of each proof is fixed cost (OT setup and the final flush). Data sizes move in steps of about 0.48 MB per direction because the OT extension runs in fixed chunks.

6.3 End to end

Live sessions against api.venice.ai (`deepseek-v4-flash`, 40 tokens), client and node on one host, medians of 5 sessions each. The single-party variant (v1, section 7) is measured on the same setup.

	v1 (single-party)	v2 (this protocol)
Cold: request start → body sent	0.73 s	0.89 s (2PC 0.55 s, of which tables 0.49 s; π_{NK} 0.17 s)
Cold: suffix proof held before U starts	0.45 s (π_{C2})	0.10 s (π_{C2S})
Cold: time to first token	1.90 s	2.43 s
Pre-warmed: arming time per session	0.71 s	0.88 s
Pre-warmed: claim → body sent	3 ms	5 ms
Pre-warmed: time to first token	1.41 s	0.90 s

The protocol's own path to U starting work is 1.18 s for v1 and 0.99 s for v2 cold, and 0.45 s against 0.10 s pre-warmed. The time-to-first-token figures are dominated by the provider's own latency, which varied by several hundred milliseconds during the runs: the cold v2 median is higher although its protocol path is shorter. An earlier interleaved comparison of the single-party variant with a direct HTTPS call (60 tokens, n = 8) measured +1.07 s to first token cold and about +0.25 s pre-warmed. π_{C3S} and settlement run after the client's stream has ended and add no latency.

Pre-warming. Because the garbled tables are single-use and N may write only in epoch 0, the preprocessing cannot be shared between requests; pre-warming is how the protocol hides it. A client arms a session before a request exists: setup and reservation, preprocessing and the joint key share. It stops there, before the ClientHello, so an armed session holds no connection to the provider (N dials U only when the ClientHello arrives). A request claims the session: the handshake with both online 2PC phases, π_{NK}, the credential record and KeyUpdate, the body and π_{C2S}. An armed session holds a reservation until it is claimed or discarded (released at no charge), and its body length is fixed at arming: the client pads a shorter body with JSON whitespace between the private part and the suffix, choosing the length from recent bodies (power-of-two buckets). The client daemon arms only while it sees traffic: after a confidential request for a given node, model and token cap it keeps one armed session (two on bursts or overlapping requests, at most three per key and four in total), retires sessions before the node's session limit, and closes them a few minutes after the last request, so an idle client holds nothing. (The reference client's `--arm` mode, used for the local benchmarks, arms further, through the handshake and π_{NK}.)

The proofs' own setup is pre-warmed too. Each proof needs OT/VOLE correlations (Chou-Orlandi base OT, KOS, Ferret), and generating them per proof costs about twelve round trips and up to 1.5 MB, which over a home uplink took 5.4 s for π_{NK} and 3.0 s for π_{C2S}. These correlations are independent of the request, so an armed session also opens one proof channel and generates, during arming, the correlations for both directions (N proves π_{NK}, C proves π_{C2S} and π_{C3S}), with a fresh Δ per session; each correlation is used once, a failed proof closes the channel, and a session without a channel falls back to one connection per proof.

Measured against a deployed node from a home connection (about 2-7 Mbit/s up): a cold request reached its first token after about 35 s, almost all of it the client uploading its half of the 2PC preprocessing. With an armed session, π_{NK} took 0.21-0.25 s, π_{C2S} 0.04 s and π_{C3S} 0.06 s over the channel, claim to body took about 1.0 s (handshake and both online 2PC phases), and the first token came after 2.0-3.2 s, most of it the provider's own latency. Arming took about 55 s on that uplink, in the background.

6.4 Over a network and bandwidth

With client and node 50 ms apart, the component figures give the following critical-path work before U starts (sums of the component benchmarks, without the relayed handshake round trips that both versions share):

	v1	v2 at 100 Mbit/s	v2 at 20 Mbit/s
Cold	3.8 s ($\pi_{C1} + \pi_N + \pi_{C2}$)	4.2 s (tables 1.8, online 2PC 0.45, π_{NK} 1.17, π_{C2S} 0.79)	11.0 s (tables 8.6; the rest as at 100 Mbit/s)
Pre-warmed	1.5 s (π_{C2})	0.8 s (π_{C2S})	0.8 s plus the slower link

Cold, v2 roughly matches v1 on a fast link and is several seconds slower on a slow one. Pre-warmed, v2 is faster, because only its cheaper suffix proof is on the path.

Data per request: v1 moves about 5.9 MB of proofs; v2 about 42 MB of 2PC traffic and 3.9 MB of proofs, 46 MB in all. N uploads about 23 MB per v2 request (100 requests per minute is about 0.3 Gbit/s), against about 3 MB for v1. The 2PC cost is constant in the size of the prompt and the response.

6.5 Live verification

Live sessions against the case-study provider completed with all proofs verified, a signed receipt passing every client check, and the cost within the reservation (63 of 2,282 msat in the example of section 4.6). They were billed through the node's ordinary accounts, with both an API key and a one-off e-cash token as the credential. Sessions over the

production SDK transport and the client daemon, driven by an off-the-shelf OpenAI client, settled; the node logged no plain chat request. Earlier runs of the same settlement path covered an upstream 400 settled at zero cost with only the response head disclosed (the provider's error body quoted the request's private field name, and no log contained it), and idle connections that never sent a setup, which caused no connection to the provider.

Adversarial runs form a regression gate (`crypto/tools/adversarial-model-swap.sh`) with an honest control and an expected outcome per case, checked against the node's billing log. On v2 it passed 6 of 6: the control settled, a decoy model in the private part of the body settled as the setup model, and every attack was refused with the reservation released:

- a suffix other than the canonical one;
- a body naming a pricier model, or a model the node does not offer, with the suffix withheld (U never received a complete request);
- a body ending in another model's suffix while claiming the canonical one (π_{C2S} fails).

A client that puts its own P-256 key share in the ClientHello instead of the joint one is refused before anything reaches the provider. The unit and integration suites of all components pass.

7. v1: the single-party variant

The protocol has a simpler variant, which we call v1, without any 2PC. A node runs one version and announces it in its signed offer; clients follow the offer.

How it works. C runs the TLS handshake alone, with an ordinary X25519 key share, and discloses only c_{hs} and s_{hs} to N, which verifies the handshake as in v2. C then sends a KeyUpdate in epoch 0, so that epoch 1 starts. For epoch-1 sequence 0, C gives N **one-shot record material**: the CTR keystream truncated to the head length, the GHASH key and the tag mask. π_{C1} proves to N, with HS as the witness, that this material comes from the session's key schedule (HS through MS, $c_{ap,0}$, $c_{ap,1}$, K_1, IV_1), so that C cannot hand out a keystream that rewrites the head around N's key. N forms its record by XOR, proves its content with π_N (the v2 π_{NK} with a public keystream), and writes it; C then sends a second KeyUpdate and writes the body in epoch 2. π_{C2} and π_{C3} prove the suffix and the usage event like their v2 counterparts, but take HS as the witness and anchor on the public handshake values, so each recomputes the key schedule (28 to 36 compressions).

What it gives up. C holds the whole epoch-1 key K_1: it needs it to seal its own KeyUpdate at sequence 1. The API key is therefore exactly $C_1 \oplus S$, and P1 holds only as long as C never sees C_1. The implementation never sends it to C, but anyone on N's egress path who colludes with C (a shared cloud network, a TLS-inspecting middlebox, the hosting provider, a CDN edge) can recover the key, and one leak exposes it until it is rotated. Short-lived keys with spend caps limit the loss without removing the assumption. In v1 P1 is an operational assumption; in v2 it is a cryptographic property. P2 to P6 are the same in both versions, and v1's residual risk 1 is narrower: N can write exactly one record of fixed length.

When v1 is preferable.

	v1	v2
Client can recover the API key	yes, if it sees N's egress ciphertext	no
New cryptography in the trust base	ZK proofs	ZK proofs and a 2PC
Key exchange	X25519	P-256
Data per request	≈5.9 MB	≈46 MB
Node upload per request	≈3 MB	≈23 MB
Cold critical-path work at 50 ms RTT	≈3.8 s (100 Mbit/s)	≈4.2 s (100 Mbit/s), ≈11 s (20 Mbit/s)
Pre-warmed suffix proof	0.45 s local, 1.5 s at 50 ms RTT	0.10 s local, 0.8 s at 50 ms RTT

v1 suits nodes whose upstream path is under their own control, clients on slow or metered links, and clients that cannot pre-warm. v2 suits nodes on shared or untrusted infrastructure and nodes with valuable keys, and is faster when sessions

are pre-warmed.

8. Deployment pattern

Confidential mode plugs into an existing routing deployment as an alternative *transport*; everything else stays the deployment's own code. Negotiation is per request and stateless:

1. **Offer.** The node publishes a signed offer (protocol version, upstream hostname, key length and alphabet, head template, pinned suffix keys, a signed rate snapshot, attestation key) at a public endpoint.
2. **Model listing.** The model listing gains one additive field per model that references the offer; old clients ignore it.
3. **Opt-in.** A client opts in per request, for `POST /v1/chat/completions` only. It routes only to nodes that advertise the field for the model, and requires that the offer signature verifies with a pinned node key (not just the key the offer names), that the upstream hostname is on its trusted hostname list (an empty list refuses every host), and that its prover is available. It rebuilds every public input of π_{NK} from the signed offer.
4. **Billing** is the node's ordinary billing (section 4.8). The client's credential, reservation, settlement, balance and refunds are the existing ones. In the reference integration the node adds a WebSocket route and an offer; the client SDK adds one transport branch beside its other transports, so credential acquisition, top-ups, failover and balance accounting are unchanged; the daemon adds a configuration block and passes the opt-in through.
5. **Response.** The client streams the decrypted body to the application as records arrive, or folds the stream into one completion object if the caller did not ask for streaming (the pinned suffix always requests streaming, because the usage event only exists there). Checks that complete after the response (the disclosure proof, the receipt, the usage, the cost, the served model) are not dropped: a failure fails a non-streaming call and ends a stream with an error.

For existing deployments the change is additive: the ordinary proxy path and old clients are untouched. Requests that opted in fail closed and are never sent on the plain path, since a silent downgrade would break the privacy promise.

9. Related work and alternatives

MPC-TLS (TLSNotary). The prover and a notary jointly act as the TLS client, and all record encryption and decryption runs inside the MPC. With the roles arranged so that the node is the notary, it gives prompt confidentiality and lets the API key be the node's private input, and it works with TLS 1.2. The cost scales with the payload and there is no streaming: we measured about 5.8 s and 41.5 MB for a 2 KB request, 34 s and 377 MB for 32 KB, and 129 s and 1.44 GB for 128 KB (100 Mbit/s, 50 ms). Our 2PC moves a similar amount of data as the 2 KB case, but it covers only the key schedule, so its cost is constant, can be precomputed, and the payload then streams at native speed. TLSNotary's key-exchange conversion is the basis of our ECtF.

DEAP and Garble-then-Prove. TLSNotary's dual execution with asymmetric privacy (DEAP) and Garble-then-Prove (Xie et al., USENIX Security 2024) obtain malicious security cheaply by having one party open its inputs, or its key shares, after the session, so that the other can check the garbling and later prove statements about the transcript. That fits a notary whose shares are worthless after the session. It does not fit our setting: N's share would reveal the master secret, hence K_0, hence the API key, which is a long-term secret. We use full dual execution, with an equality check on output labels (Huang, Katz and Evans 2012), in which nothing is ever opened, at the price of garbling every circuit twice and accepting one bit of leakage.

DiStefano (Celi et al., 2023) runs the TLS 1.3 handshake as a 2PC with authenticated garbling, which is maliciously secure without leakage but needs heavier preprocessing. It could replace our dual execution where the leakage bit matters more than bandwidth.

DECO (Zhang et al., ACM CCS 2020) introduced the three-party handshake, in which prover and verifier secret-share the session keys so that the verifier is convinced of data the prover received. Its goal is to convince the verifier of provenance; ours is the reverse split, in which each party receives a different epoch's key in full and the payload never enters the MPC.

Proxy zkTLS attestors (the "verified mode" deployed on Routstr nodes; Reclaim-style attestors). The node is the TLS client and an attestor checks its transcript. This gives request and response integrity and portable proofs, but the node and the attestor see the plaintext; we measured +10.8 s to first token on a live node.

TEE (enclave) designs. The node runs in a hardware enclave that holds the API key and processes the prompt in the clear, and users verify attestations. This is fast and streams, and it works with any provider. It rests on the hardware vendor and on the measured code, and the guarantee is only as good as the user's review of every software version the enclave runs (section 1).

The two approaches compose. Our protocol removes the need to trust the reseller in the middle, while C remains the TLS endpoint of the provider. If the provider itself serves inference from an enclave and binds its TLS key or an encryption key to an attestation (attested or end-to-end-encrypted inference offered by some providers), C can verify that attestation end to end through N, which only relays sealed records. The user then gets math-based guarantees about the node and the provider's attestation about the provider, without having to review the node's software.

	This work (v2)	v1 variant	MPC-TLS	Proxy zkTLS	TEE
Prompt hidden from node	yes	yes	yes	no	yes
API key hidden from client	yes	if N's egress is private	yes	yes	yes
Trusted hardware or attested software	no	no	no	no	yes
Streaming, native-speed payload	yes	yes	no	yes	yes
Added latency, pre-warmed (local)	0.10 s proof	0.45 s proof	5.8-129 s	≈10 s	≈0
Data per request	≈46 MB, constant	≈5.9 MB	41 MB-1.4 GB	transcript size	≈0
Third-party-verifiable transcript	no	no	yes	yes	attestation only
Provider requirement	TLS 1.3 profile with P-256	TLS 1.3 profile	TLS 1.2+	TLS 1.2/1.3	any

The approaches compose: proxy zkTLS provides integrity for unmodified providers, MPC-TLS covers providers that fail the profile, and our design is the fast lane where a provider's TLS and JSON semantics cooperate.

10. Conclusion

We presented a protocol that gives LLM routing users prompt confidentiality, request and response integrity, and upstream-anchored accounting against a node that relays ciphertext, and gives the node a guarantee that its API key stays hidden from the client, with no trusted hardware and no MPC over the payload. Its trust rests on the provider's TLS certificate and on cryptography, not on attested software. The mechanisms are a two-party TLS key schedule that gives the node the first application epoch and the client every other one, a proven public suffix for parameter pinning, and a proven usage disclosure for settlement. A complete implementation, verified against a live provider, reaches the provider 5 ms after a pre-warmed request is claimed, plus a 0.10 s suffix proof; its cost is about 21 MB of precomputable garbled tables in each direction per request. The single-party variant needs a tenth of the bandwidth where the node can keep its egress private.

Appendix A. Case study: the provider profile in practice

A.1 Verified behaviours (live gate `crypto/tools/provider-profile-check.py`, checks A1-A11, 11/11 on 2026-10-10; rate limits, 5xx and timeouts count as inconclusive, never as a pass, and each rejection check asserts the exact status and error text against a valid control).

- **Transport:** a client offering only TLS 1.3 with `TLS_AES_128_GCM_SHA256` is served; a client KeyUpdate sent in the middle of the request is accepted and the server answers completely and closes (`Connection: close`, HTTP/1.1). The v2 live sessions establish that a single secp256r1 key share is accepted without a HelloRetryRequest.
- **Waits for the full body:** nothing for 8 s with the body incomplete, then a response within 0.7 s of the last bytes.
- **Last-wins:** a bad model then a good one gives 200; a good one then a bad one gives 404; a `max_tokens` of 200 followed by 8 behaves like 8.
- **Replace, not merge:** a later `venice_parameters` object replaces an earlier one; an invalid `character_slug` that is overridden is ignored.
- **Strict parsing (sampled):** an unknown top-level field and malformed JSON are rejected with 400, including a suffix landing in an open object or string. The 400 body for an unknown field quotes the field's name, which is why an error disclosure covers only the response head (section 4.8).
- **Token cap:** `max_completion_tokens` caps output including reasoning, with a small overshoot on tiny caps (3 billed up to 6, 40 billed 39). The reservation covers this easily.
- **Input bound:** with the pinned `include_venice_system_prompt: false` a 224-byte request billed 6 prompt tokens. Without the pin the provider injects about 1.6k tokens of system prompt.
- **Unpinned price fields (sampled):** `service_tier`, `web_search_options` and `prediction` are rejected.
- **Images:** every offered model rejects image parts, both `image_url` and data URIs (`crypto/tools/regression-modality.py`, 8/8).
- **Usage event:** the terminal event carries usage, cost and model with empty `choices`, followed by `[DONE]`.
- **Fake usage events:** a prompt instructing the model to emit them produced none. Every imitation was escaped inside a JSON string, and the raw stream contained no CR/LF inside events.

A.2 Profile application. The pinned suffix for this provider: `,"model":M,"max_tokens":T,"max_completion_tokens":T,"n":1,"` with `max_completion_tokens` pinned because it overrides `max_tokens` (a bypass observed in testing) and the provider-specific parameters pinned to keep web search and system-prompt injection off the bill.

A.3 Other providers. The generic pieces (epochs, 2PC, proofs) port unchanged. The open questions per provider are profile items 4 to 6 and 8, plus whether HTTP/1.1, P-256 and a mid-request KeyUpdate are accepted (some CDNs pin HTTP/2). Where a provider fails, the fallbacks are MPC-TLS for confidentiality or the existing integrity-only path. Per-provider assessments are in the project's `docs/ADVERSARIAL-QA.md`.

References

- S. Celi, A. Davidson, H. Haddadi, G. Pestana, J. Rowell. *DiStefano: Decentralized infrastructure for sharing trusted encrypted facts and nothing more.* NDSS 2025 (ePrint 2023/1063).
- Y. Huang, J. Katz, D. Evans. *Quid-Pro-Quo-tocols: Strengthening semi-honest protocols with dual execution.* IEEE S&P 2012.
- M. Keller, E. Orsini, P. Scholl. *Actively secure OT extension with optimal overhead* (KOS). CRYPTO 2015.
- P. Mohassel, M. Franklin. *Efficiency tradeoffs for malicious two-party computation.* PKC 2006.
- E. Rescorla. *The Transport Layer Security (TLS) Protocol Version 1.3.* RFC 8446, 2018; test vectors RFC 8448.
- TLSNotary project. *mpz* and *mpz2* (garbling, oblivious transfer, VOLE-based ZK) and the MPC-TLS protocol documentation.
- R. Weiss, D. Ayzenshteyn, G. Amit, Y. Mirsky. *What was your prompt? A remote keylogging attack on AI assistants.* USENIX Security 2024.
- X. Xie, K. Yang, X. Wang, Y. Yu. *Lightweight authentication of web data via garble-then-prove.* USENIX Security 2024.

- K. Yang, P. Sarkar, C. Weng, X. Wang. *QuickSilver: Efficient and affordable zero-knowledge proofs for circuits and polynomials over any field.* ACM CCS 2021.
- S. Zahur, M. Rosulek, D. Evans. *Two halves make a whole* (half-gates). EUROCRYPT 2015.
- F. Zhang, D. Maram, H. Malvai, S. Goldfeder, A. Juels. *DECO: Liberating web data using decentralized oracles for TLS.* ACM CCS 2020.